

Scalable AI Business Model

GLOSSARY 03 - SHEET 6 / 8

A model whose marginal cost of serving an additional user does not rise with volume

PROFESSIONAL DEFINITION

A scalable AI business model is one in which the cost of serving an additional customer or processing an additional transaction does not rise in proportion to volume, so that value delivered grows faster than the resources consumed. Iansiti and Lakhani identify three properties that distinguish such models - the absence of operational diseconomies of scale, positive returns to scope from reusing data across domains, and continuous learning effects - and locate the constraint on all three in the operating model rather than the algorithm (Iansiti and Lakhani, 2020).

WHO DEFINED IT ORIGINALLY

Marco Iansiti and Karim R. Lakhani of Harvard Business School developed the account in *Competing in the Age of AI* (2020), extending their earlier work on digital platforms and drawing on Arthur's (1996) analysis of increasing returns and on Parker, Van Alstyne and Choudary's (2016) platform economics.

WHY THIS DEFINITION WAS DEVELOPED

Traditional firms scale by adding people, and their operating models therefore encounter diseconomies as coordination costs rise. Iansiti and Lakhani needed to explain why digitally native competitors did not, and located the difference in an architecture where the software, not the workforce, absorbs additional volume - which reframes the strategic question as one of operating-model design.

FIGURE 1 - THREE PROPERTIES OF A SCALABLE MODEL

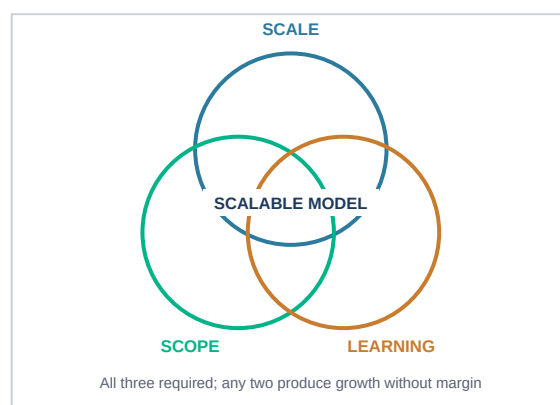


Figure 1. Scale without learning yields a large business that does not improve; learning without scale yields an improving business that cannot grow. The binding constraint on all three is the operating model, not the algorithm (after Iansiti and Lakhani, 2020).

HOW IT IS USED

Test each of the three properties against the specific model: whether marginal delivery cost is flat, whether the data collected is reusable in adjacent domains, and whether use improves the system; a model failing all three will not scale regardless of the technology used.

WHEN IT IS USED

At business-case approval for an AI-enabled product, when unit economics deteriorate as volume rises, when assessing a competitor's growth trajectory, and when a services business proposes an AI product.

BY WHOM IT IS USED

Product and business-model owners, chief financial officers testing unit economics, venturing and corporate development teams, and boards assessing whether growth will be profitable.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

The predecessor concept was economies of scale from manufacturing economics, which assumed a fixed asset base amortised over volume and produced diminishing returns beyond an optimum. The AI-era account substitutes learning effects and scope economies, in which the asset improves rather than depreciates with use. Its important qualification is that these properties belong to the operating model: a firm can hold excellent models and still fail to scale if humans remain in the delivery path.

RELATED DEFINITIONS IN THIS COURSE

Business Model Innovation - Unit 1

AI Operating Model - Unit 3

Profit Pool - Unit 3

Competitive Advantage - Unit 3

Value Migration - Unit 1

SOURCES (HARVARD REFERENCING STYLE)

Arthur, W.B. (1996) 'Increasing returns and the new world of business', *Harvard Business Review*, 74(4), pp. 100-109.

Iansiti, M. and Lakhani, K.R. (2020) *Competing in the Age of AI: Strategy and Leadership When Algorithms and Networks Run the World*. Boston, MA: Harvard Business Review Press.

Parker, G.G., Van Alstyne, M.W. and Choudary, S.P. (2016) *Platform Revolution*. New York: W.W. Norton.